Melis Kilic

AI and Data Engineer

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About Me

AI and Data Engineer building the infrastructure that makes AI features measurable and business data trustworthy. Works end to end, from agentic workflows and their evaluation down to the multi-tenant database rules that keep one customer's data out of another's reports, delivering resilient systems stakeholders can rely on. Seeking a role where production reliability, data correctness and measurable quality carry the same weight as feature delivery.

Professional Experience

AI and Data Engineer, QKare Information Technologies, Sivas

2025 to Present

B2B HR analytics SaaS platform, plus a TÜBİTAK 1507 funded R&D programme building AI recruiting modules. Work spans agentic AI services, governed data layers, evaluation pipelines and FastAPI backend infrastructure.

Multi-Tenant HR Analytics Platform

- Shipped multi-agent analytics and coaching features to production on LangGraph and FastAPI, turning workforce data into conversational insights HR teams use daily.
- Architected a shared FastAPI gateway for a monorepo of AI services, with authentication, request-traceable JSON logging, OpenTelemetry hooks, project scaffolding and multi-stage Docker builds built in.
- Replaced subjective quality review of a conversational AI product with an automated LLM-as-judge harness scoring simulated multi-turn sessions on 5 weighted dimensions.
- Resolved aggregation errors skewing customer reporting by modeling 56 mart tables through a 109-model dbt Core layer, then converting the underlying rule into a test that runs on every pipeline change.
- Isolated every customer's data in a shared platform by scoping database calls to the requesting user with PostgreSQL row-level security (RLS), backed by tests asserting no query path crosses tenants.

Luna AI — TÜBİTAK 1507 funded R&D programme

- Engineered an LLM CV extraction pipeline to a 99.58% evidence-grounded rate across 239 profiles, screening every document through an offline prompt-injection and document-type guard before any model call.
- Designed the canonical candidate-profile schema every module builds against, published as a versioned internal package, requiring each extracted field to cite its source span so a fabricated value fails validation before reaching a reviewer.
- Overhauled CV anonymization around a locally hosted LLM detector and a mandatory output audit, so personally identifiable information (PII) never reaches the cloud extraction model.

AI Research Intern, Inncrea Software and IT, Sivas

Aug 2024 to Sep 2024

Research internship on a medical imaging study that continued as an open-source project after it ended.

Stroke Classification — TEKNOFEST Healthcare AI, Hugging Face

- Made model comparisons trustworthy by fixing the evaluation protocol before the models: stratified folds, class balancing and identical augmentation across every backbone tested.
- Outperformed a heavier InceptionV3 teacher by distilling it into a compact EfficientNet-B0 student at 98.0% F1, with a soft-voting ensemble reaching 98.2% accuracy over 49,500 CT images.
- Held performance on a completely external CT dataset with different scanner calibrations, showing the distilled model generalized beyond its own training distribution.
- Published the weights, dataset and a live demo on Hugging Face, later deploying the model to Google Cloud Run as a FastAPI and ONNX Runtime service with per-prediction logs streamed into BigQuery.

Education

Sivas Cumhuriyet University, Bachelor's Degree, Computer Engineering, Turkey — 2021 to 2025

AGH University of Krakow, Erasmus+ Exchange, Krakow, Poland — 2023 to 2024

National Technology Academy, AI Specialization Program — Dec 2024 to Jun 2025

Skills

AI and LLM Systems:	LangGraph, LangChain, RAG, Agentic AI, MCP, Prompt Engineering, OpenAI API, Claude API
Data Engineering:	SQL, dbt Core, PostgreSQL, Row-Level Security (RLS), Data Quality, ETL/ELT, Apache Superset
Evaluation:	LLM-as-Judge, OpenAI Evals, Observability, Langfuse, OpenTelemetry, MLOps
Backend and API:	Python, FastAPI, Pydantic, REST APIs, WebSocket, Redis
Cloud and DevOps:	Google Cloud Platform (GCP), Cloud Run, BigQuery, Docker, CI/CD, GitHub Actions, GitLab CI/CD
ML and Practice:	PyTorch, scikit-learn, Hugging Face, Git, Agile / Scrum, pytest

Projects

Agentic RAG, Open Source, GitHub 2026

- Built a self-auditing knowledge assistant over documents and SQL on a 3-agent LangGraph loop (Researcher, Analyst, Auditor) that catches and retries a hallucinated answer before it reaches the user.
- Isolated every database credential in a separate Model Context Protocol (MCP) subprocess the agent cannot read, keeping SQL execution behind a read-only boundary.
- Put destructive SQL behind human approval by pausing the agent mid-run with a LangGraph interrupt and resuming from a Redis checkpoint once a reviewer decides.

Analytics Copilot, Open Source, GitHub 2026

- Designed a dbt-first natural-language analytics interface over 99K+ e-commerce orders where every aggregation lives in 6 pre-tested mart tables, so the model queries verified numbers instead of inventing a metric at runtime.
- Enforced safe query execution at 3 independent layers with a sqlglot abstract syntax tree (AST) validator that routes failures back into a LangGraph retry loop, backed by a read-only PostgreSQL role.
- Opened the same dbt foundation to business users through a separate pre-aggregated Apache Superset modeling layer, embedded behind a guest-token endpoint limited to a read-only database role.

Additional Information

Languages: Turkish (Native), English (Professional Working Proficiency)

Certificates: HCCDA-AI (Huawei Certified Developer Associate – AI), Huawei